Application of perturbation theory to hydrogenic atoms subject to internal or external electromagnetic fields :

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$a_o = 4\pi\epsilon_0\frac{\hbar^2}{\mu e^2} = 0.5293\text{Å}$

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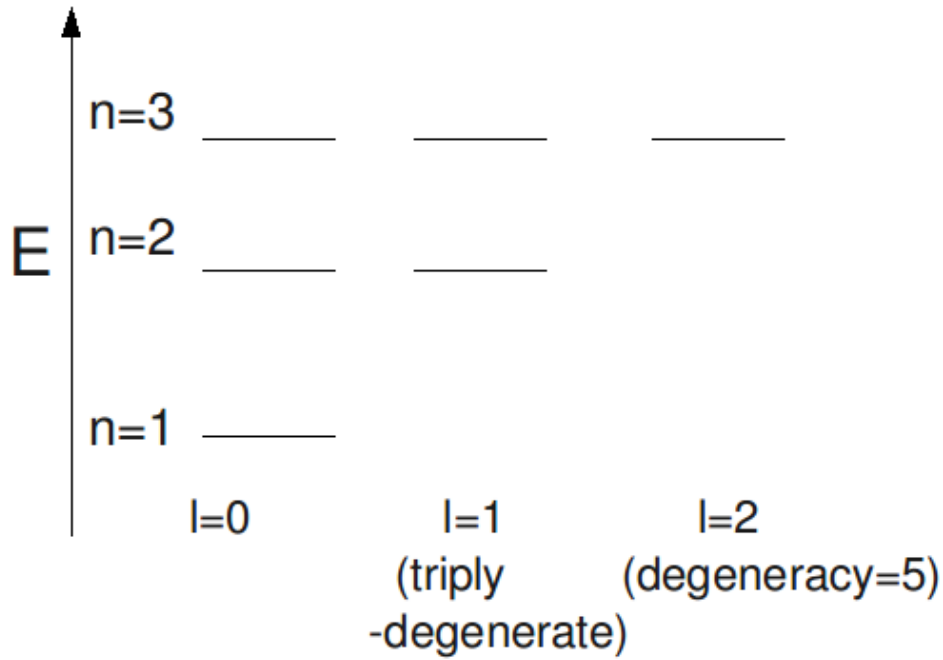
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$\therefore$ additional term in Hamiltonian

$$H' = -\frac{(p^2)^2}{8m^3c^2} = -\frac{1}{2mc^2}\left(\frac{p^2}{2m}\right)^2 = -\frac{1}{2mc^2}\left(H_0 + \frac{Ze^2}{4\pi\epsilon_0 r}\right)^2$$

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Hamiltonian matrix for a spinning charged particle, at rest in a magnetic field $\vec{B}$:
$\mathrm{H} = -\gamma \vec{B} \cdot \vec{\mathrm{S}}$
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Constants a and b are determined by initial conditions: $\chi(0) = \begin{pmatrix} a \\ b\end{pmatrix}$; $|a|^2 + |b|^2 = 1$

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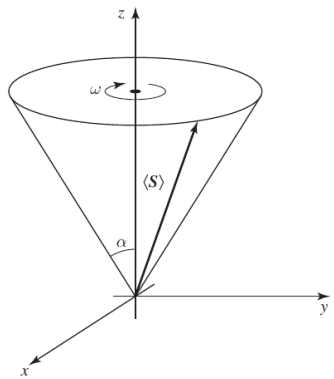
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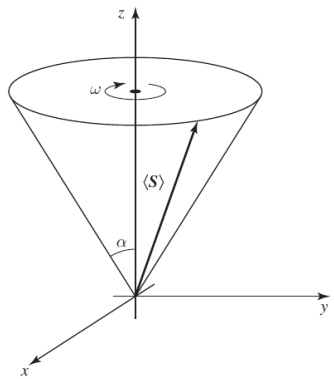
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No surprise: Ehrenfest theorem guarantees - $\langle \hat{S} \rangle$ evolves according to classical laws

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and the beam is deflected up or down, in proportion to the z component of the spin angular momentum

Classically we expect a smear, but in fact the beam splits into $2s + 1$ separate streams, beautifully demonstrating the quantization of angular momentum.
For silver atoms, all inner electrons are paired and their angular momenta cancel - net spin is simply that of the outermost—unpaired electron, so $s = 1/2$ and beam splits in two

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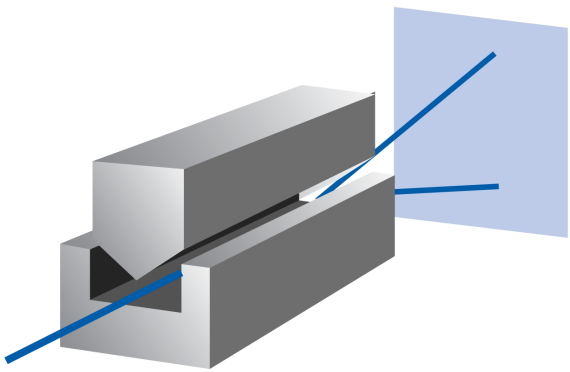
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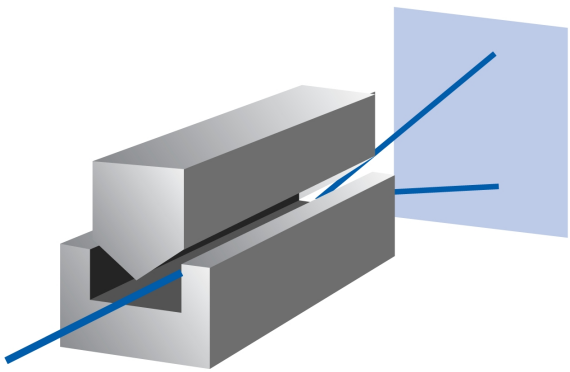


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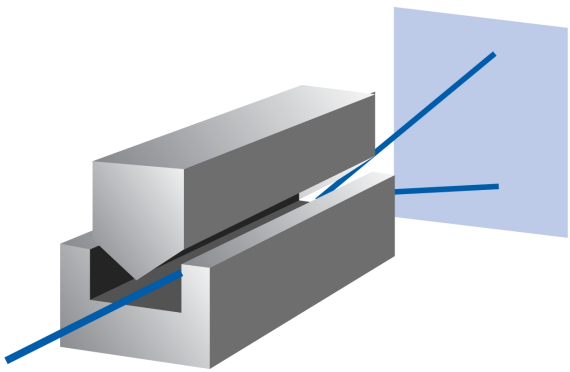
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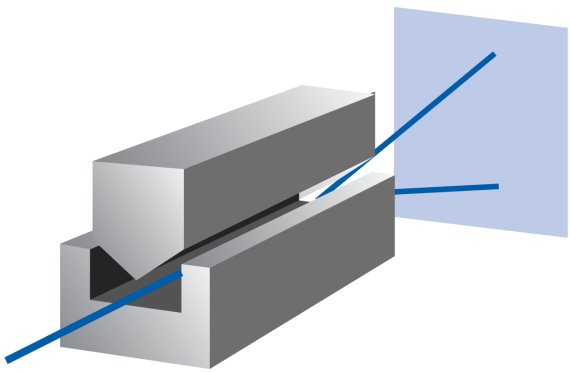
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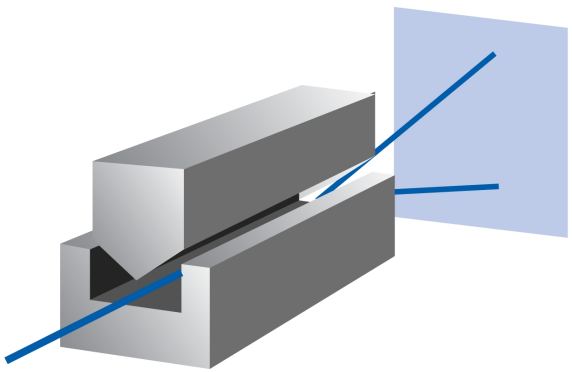
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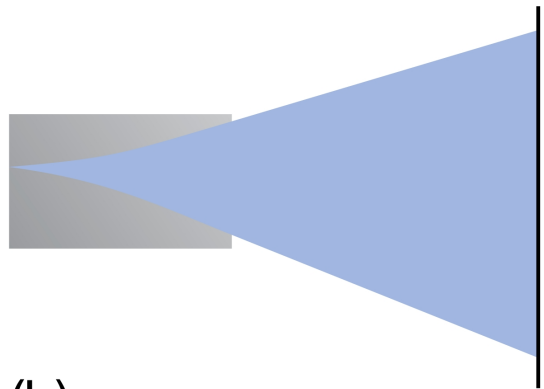
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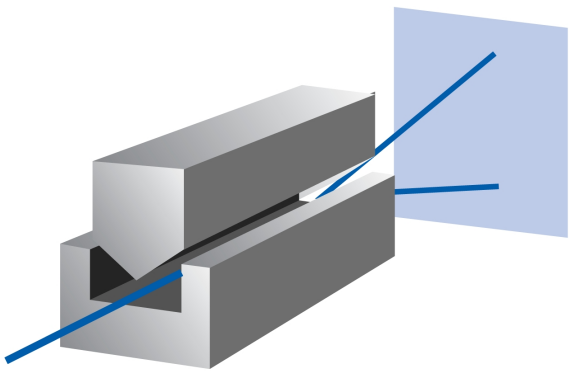


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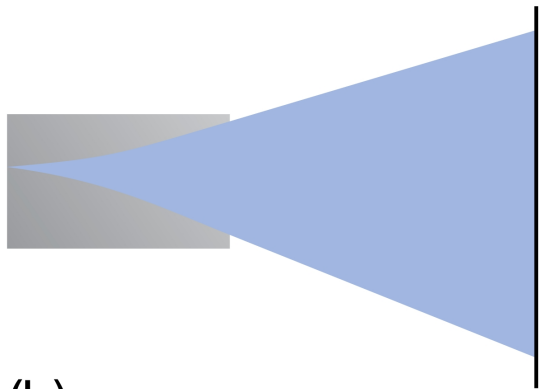
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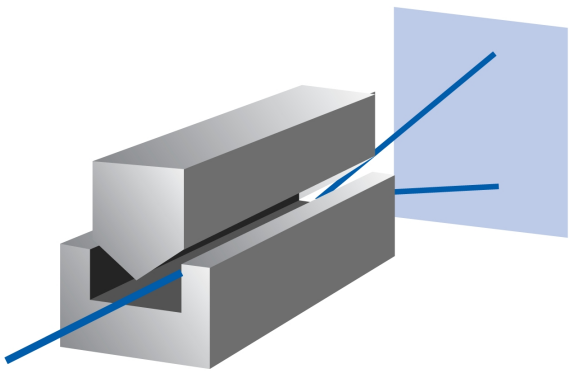
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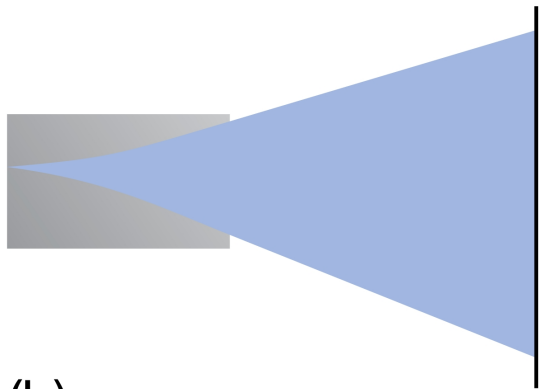
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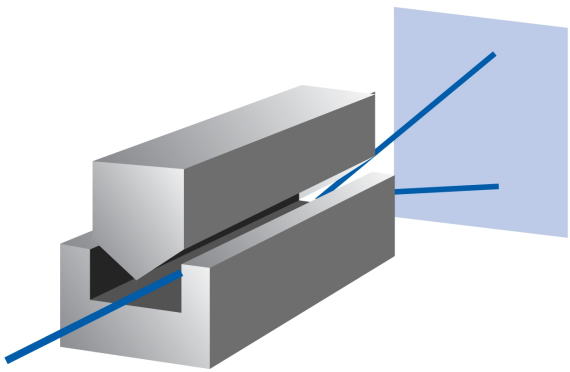
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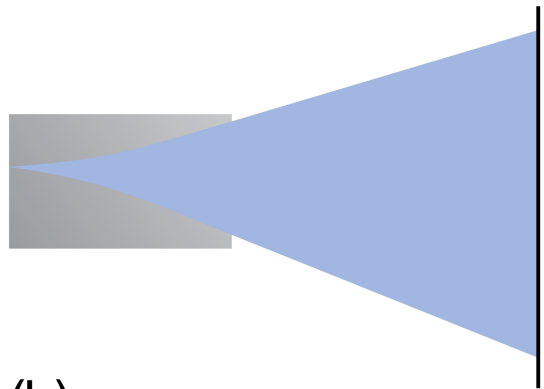
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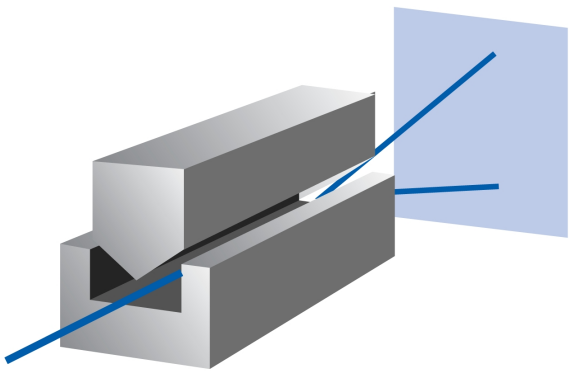
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magnetic moment quantized : 2 peaks!

Classically we expect a smear, but in fact the beam splits into $2s + 1$ separate streams, beautifully demonstrating the quantization of angular momentum.

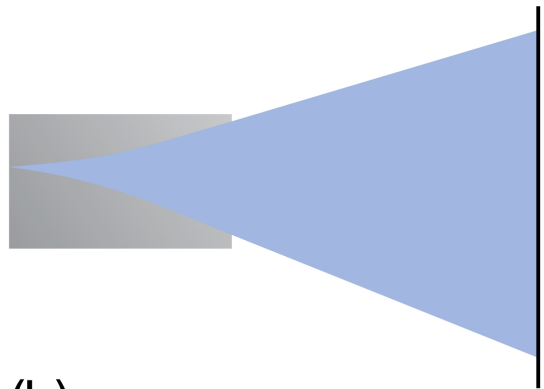
For silver atoms, all inner electrons are paired and their angular momenta cancel - net spin is simply that of the outermost—unpaired electron, so $s = 1/2$ and beam splits in two

Stern-Gerlach expt :
hot atoms 'beamed' from an oven through magnetic field which deflected atoms



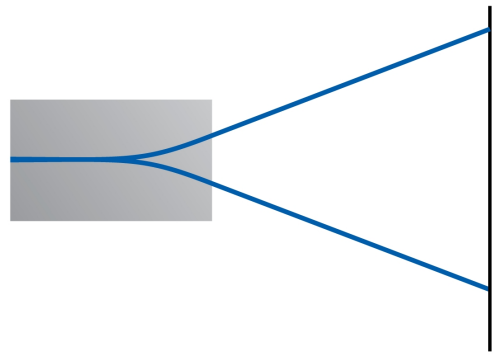
(a)

classically expected result
randomly oriented atoms
continuous distribution



(b)

actual result
Surprise
magnetic moment quantized : 2 peaks!



(c)

Back to H atom :
Magnetic Field of the Proton

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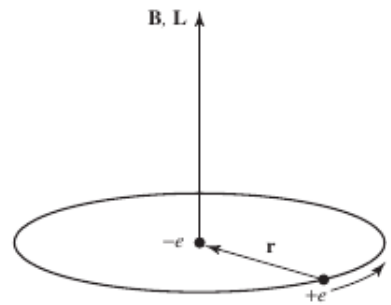
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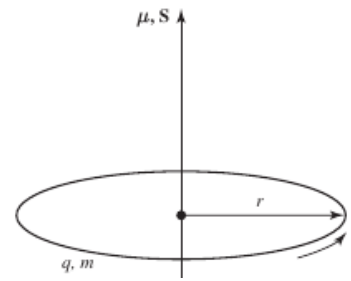
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Directions of $\vec{\mu}$ and $\vec{S}$ are the same (or opposite for electron),
$\vec{\mu} = \frac{q}{2m}\vec{S} \underset{\text{e}}{\overset{\text{for}}{=}} -\frac{e}{2m}\vec{S}$



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The quantity $\mu_B = \frac{e\hbar}{2m_e}$, is called Bohr magneton $= 9.274 \times 10^{-24}$ JT^{-1}

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- spin makes an extra contribution to magnetic moment of H atom, in addition to contribution due to orbital motion

complete Hamiltonian

$$\therefore H'_{\text{S.O.}} = \frac{Ze^2}{2m^2c^2} \cdot \frac{1}{r^3} \overrightarrow{L} . \overrightarrow{S}$$

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- singular at $r = 0$ (nucleus) unless $\overrightarrow{L} . \overrightarrow{S}$ goes to zero faster than r^3 as $r \to 0$

$$\vec{J} = \vec{L} + \vec{S}$$

$$\begin{aligned}
[\hat{J}_i,\ \hat{\vec{L}}.\hat{\vec{S}}] &= \textstyle\sum_j [\hat{J}_i,\ \hat{L}_j \hat{S}_j] = \sum_j \{[\hat{J}_i,\ \hat{L}_j]\hat{S}_j + \hat{L}_j[\hat{J}_i, \hat{S}_j]\} \\
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&= \textstyle\sum_j \{i\hbar\epsilon_{ijk}\hat{L}_k\hat{S}_j + i\hbar\epsilon_{ijk}\hat{L}_j\hat{S}_k\} = 0
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Since, $\vec{L}.\vec{S}$ is scalar invariant under rotation, $\vec{J}$ should commute with it.

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Thus we can look for simultaneous eigen states of $\hat{H}$, $\hat{J}^2$, $\hat{J_z}$, $\hat{L}^2$ when spin orbit interaction is present (i. e., eigen function of definite quantum numbers j, M, l).

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consider $l \to 0$.

$$E^{(1)} = l.\left\langle\frac{1}{r^3}\right\rangle_{n,\ l\to0} = \frac{Z^3}{n^3a_0^3}\frac{l}{l.\left(l+\frac{1}{2}\right).(l+1)}\bigg|_{l\to0} = \frac{Z^3}{n^3a_0^3}\frac{1}{\left(l+\frac{1}{2}\right).(l+1)}\bigg|_{l\to0} = \frac{2Z^3}{n^3a_0^3}$$

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&= \frac{Z^4}{2n^3}\left(\frac{e^2}{\hbar c}\right)^4.\mu c^2\,\frac{1}{(2l+1)(l+1)} \\
&= \frac{(Z\alpha)^4}{2n^3}.\mu c^2\,\frac{1}{(2l+1)(l+1)}
\end{aligned}
$$

$$\therefore E^{(1)}_{n,\,l,\,j=l+\frac{1}{2}} = \frac{Ze^2}{2\mu^2c^2}\,\frac{Z^3}{n^3a_0^3}\,\frac{\hbar^2}{2}\,\frac{2l}{l(2l+1)(l+1)}$$

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Similarly , $E^{(1)}_{n,\,l,\,j=l-\frac{1}{2}} = -\frac{(Z\alpha)^4}{2n^3}\mu c^2\,\frac{1}{l(2l+1)};\;\; l \neq 0$

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In both relativistic and spin correction, accidental degeneracy of orbitals of different l value is split and new eigenstates (perturbed) depend on l.

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$$\therefore E_{n,\,l,\,j=l\pm\frac{1}{2}} = E_n^{(0)} - \frac{(Z\alpha)^4}{2n^4}\mu c^2\left[\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\right] = -\frac{13.6\text{eV}}{n^2}\left[1 + \frac{\alpha^2}{n^2}\left(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\right)\right]$$

first order shift,

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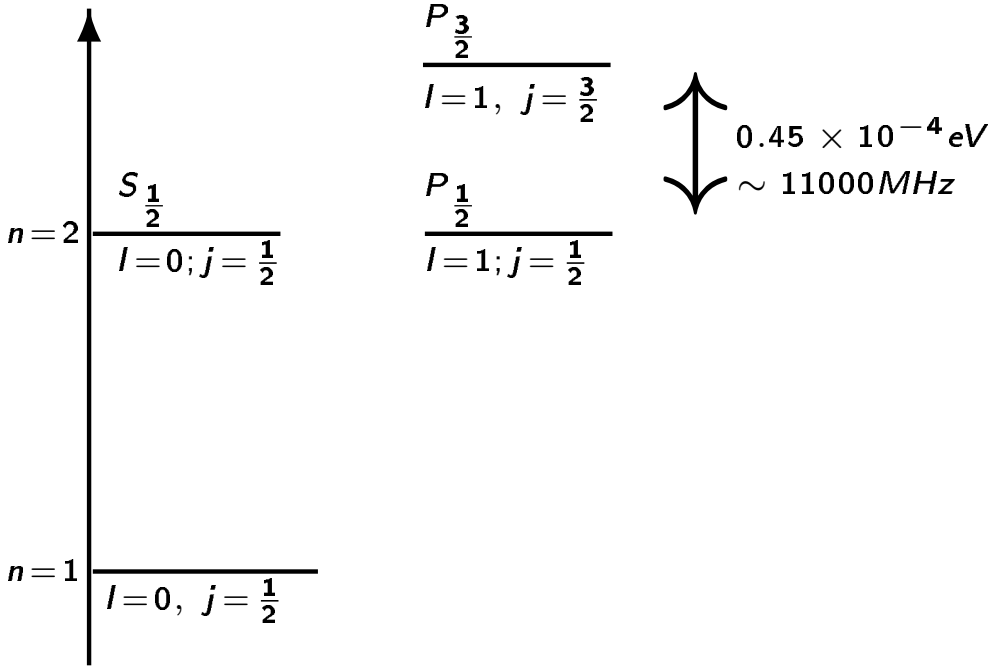
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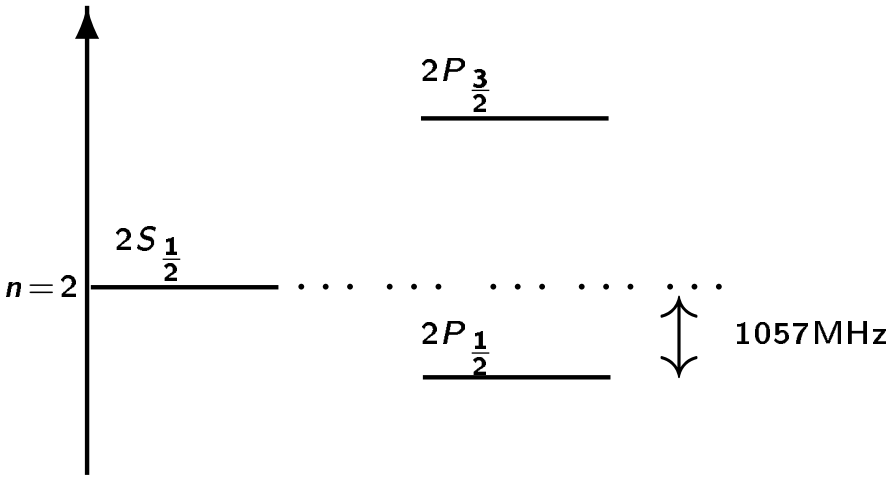
first order shift, $E_{\text{fs}} = \frac{E_n^2}{2mc^2}\left(3 - \frac{4n}{j+\frac{1}{2}}\right)$ depends only on j

energy diagram for $Z = 1$ (hydrogen) looks like the following:

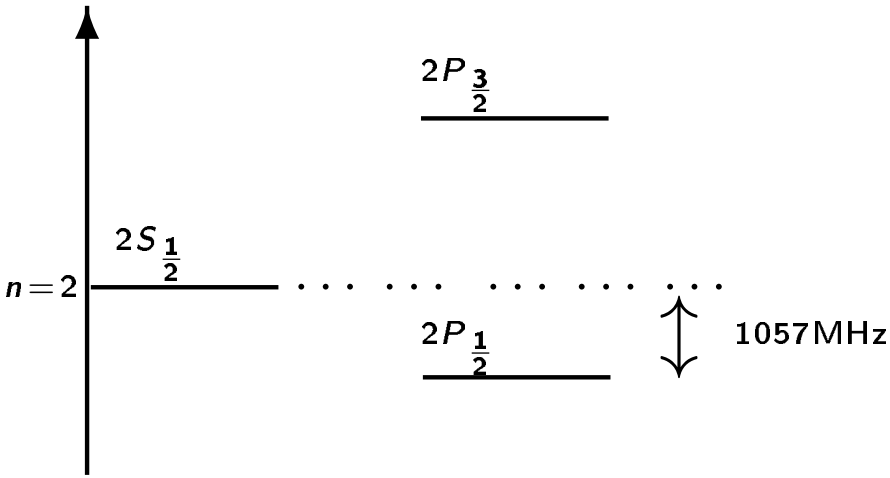


The reality is a little different for l=1 states

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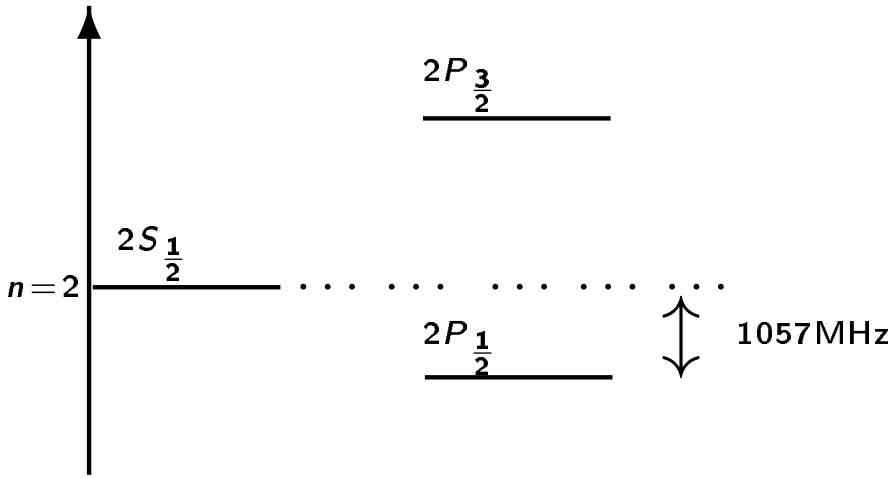


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Gap between $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$ is 1057 MHz

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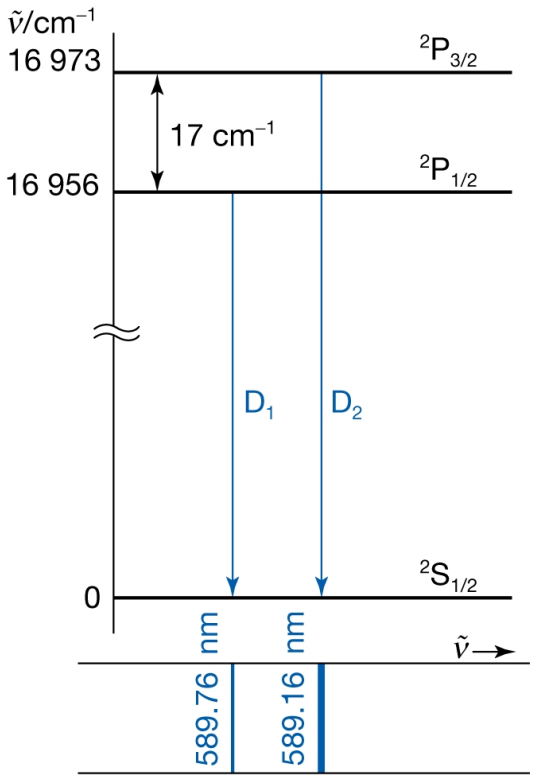


Gap between $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$ is 1057 MHz

Lamb shift :$2S_{\frac{1}{2}}$ is raised by 1040 MHz and $2P_{\frac{1}{2}}$ is lowered by 17MHz.

Sodium D lines: simple explanation:

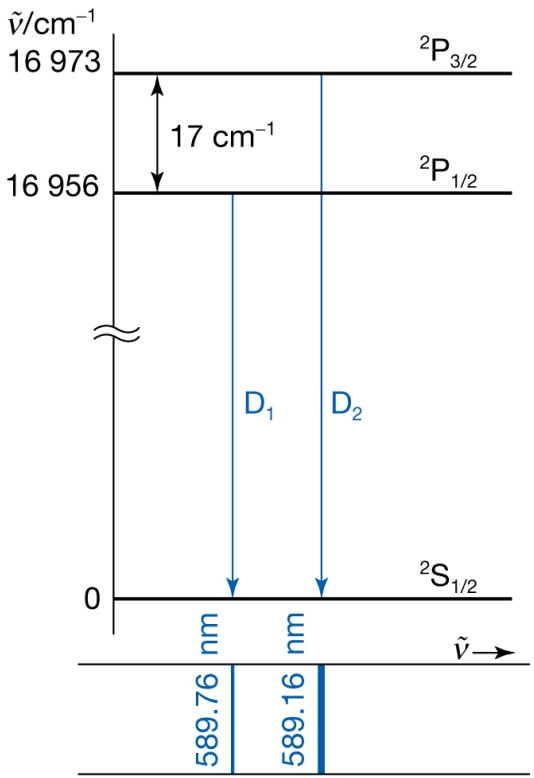
single electron: $l = L$



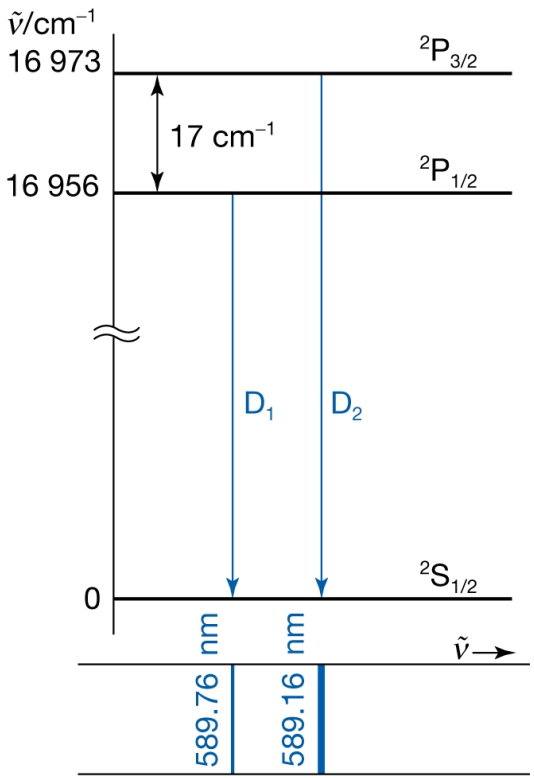
Sodium D lines: simple explanation:

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ground state: $l = 0$, $s = \frac{1}{2}$, $j = \frac{1}{2}$



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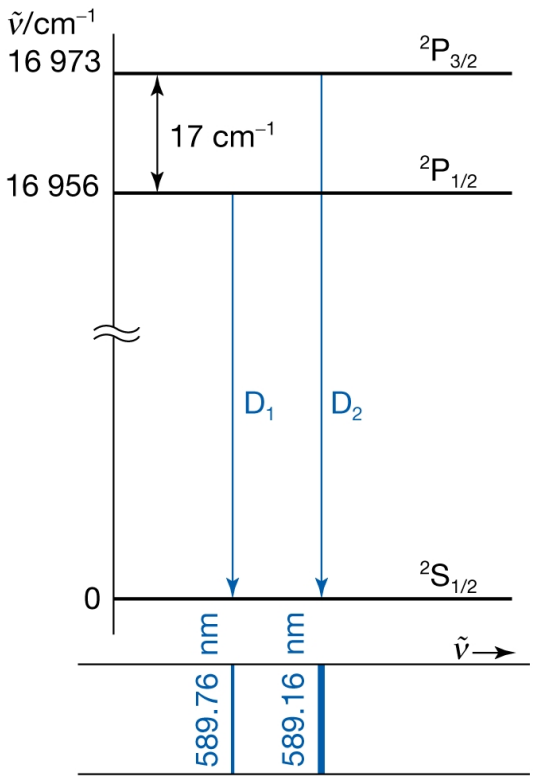


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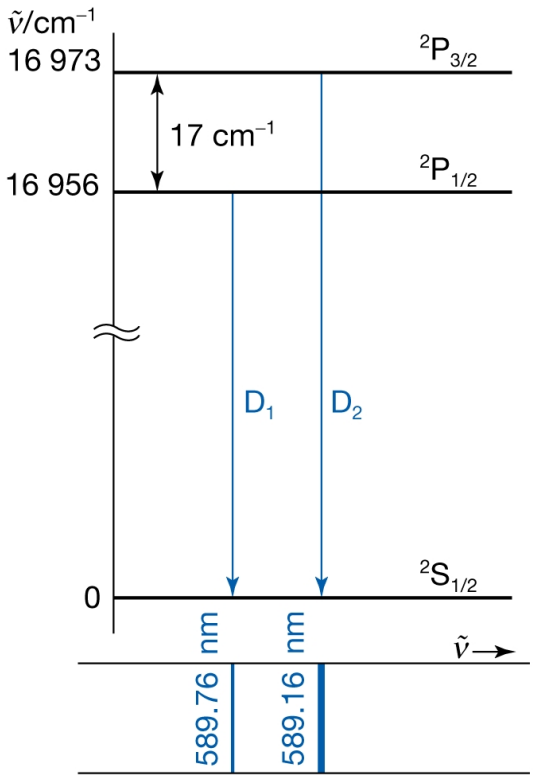


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$\langle j\ l\ s|\widehat{\vec{l}.\vec{s}}|j\ l\ s\rangle =$

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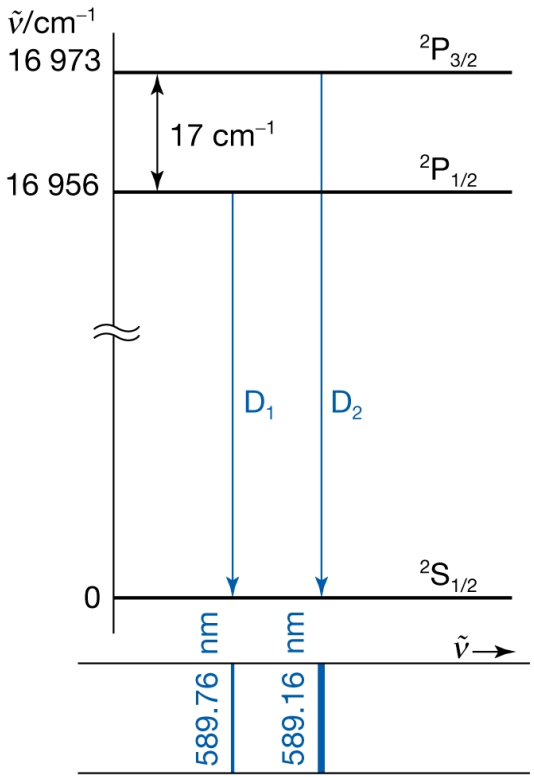


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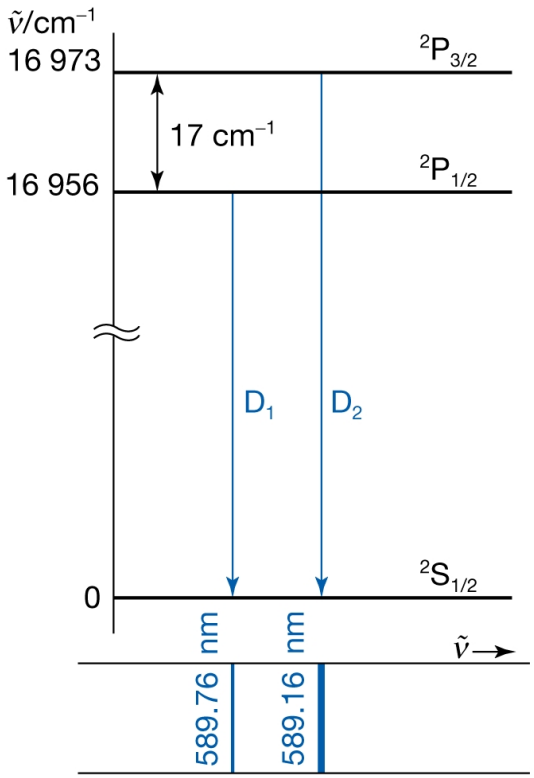


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$$\langle j\ l\ s|\widehat{\vec{l}.\vec{s}}|j\ l\ s\rangle =$$
$$= \frac{1}{2}\langle j\ l\ s|\widehat{j^2} - \widehat{l^2} - \widehat{s^2}|j\ l\ s\rangle$$
$$= \frac{\hbar^2}{2}\left[j(j+1) - l(l+1) - s(s+1)\right]$$

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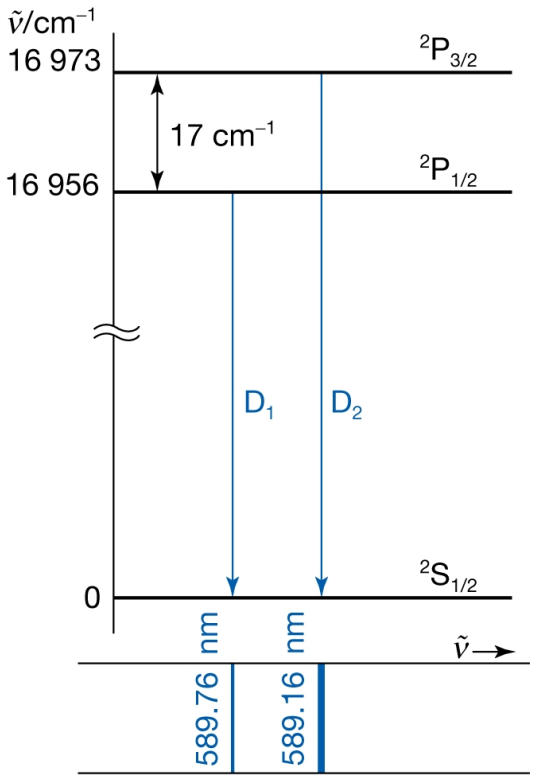
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For $j = \frac{3}{2}$, $\Delta E_{\frac{3}{2}} = \frac{1}{2}hcA$

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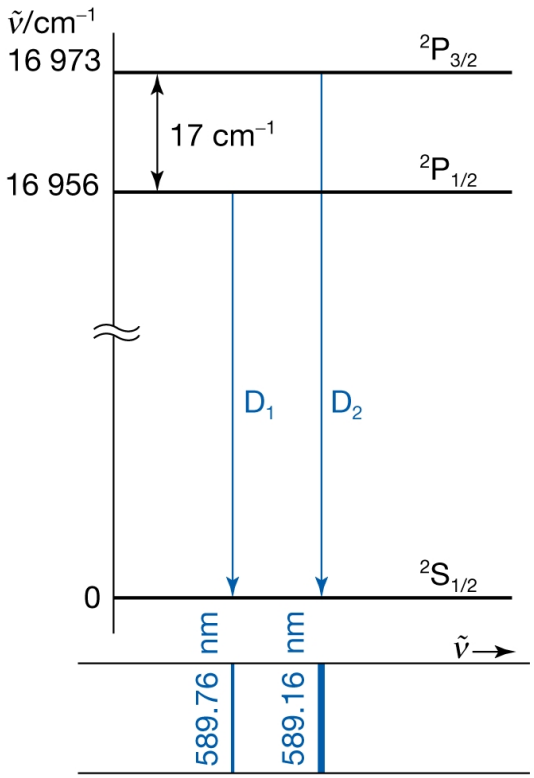
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$$= \tfrac{1}{2}\langle j\ l\ s|\widehat{j^2} - \widehat{l^2} - \widehat{s^2}|j\ l\ s\rangle$$
$$= \frac{\hbar^2}{2}\left[j(j+1) - l(l+1) - s(s+1)\right]$$

For $j = \frac{3}{2}$, $\Delta E_{\frac{3}{2}} = \frac{1}{2}hcA$
For $j = \frac{1}{2}$, $\Delta E_{\frac{3}{2}} = -hcA$

Sodium D lines: simple explanation:



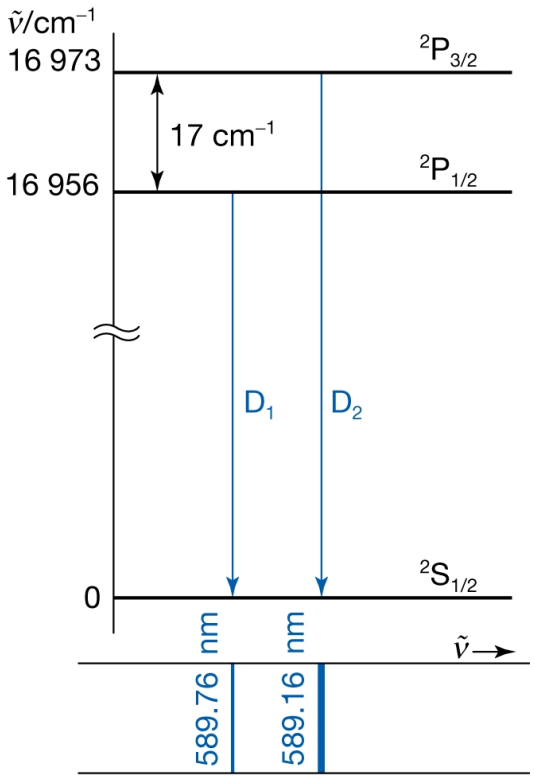
single electron: $l = L$

ground state: $l = 0,\ s = \frac{1}{2},\ j = \frac{1}{2}$
excited state: $l = 1,\ s = \frac{1}{2},\ j = \frac{1}{2},\ \frac{3}{2}$

$$\langle j\ l\ s|\widehat{\vec{l}.\vec{s}}|j\ l\ s\rangle =$$
$$= \tfrac{1}{2}\langle j\ l\ s|\widehat{j^2} - \widehat{l^2} - \widehat{s^2}|j\ l\ s\rangle$$
$$= \frac{\hbar^2}{2}\left[j(j+1) - l(l+1) - s(s+1)\right]$$

For $j = \frac{3}{2},\ \Delta E_{\frac{3}{2}} = \frac{1}{2}hcA$
For $j = \frac{1}{2},\ \Delta E_{\frac{3}{2}} = -hcA$
$\Delta\bar{\nu} = \frac{3}{2}A = 17.2\text{cm}^{-1}$

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$\therefore A = 11.5\text{cm}^{-1}$